```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    'Height': [170, 165, 180, 175, 160, 172, 168, 177, 162, 158],
    'Weight': [65, 59, 75, 68, 55, 70, 62, 74, 58, 54],
    'Age': [30, 25, 35, 28, 22, 32, 27, 33, 24, 21],
    'Gender': [1, 0, 1, 1, 0, 1, 0, 1, 0, 0]  # 1 = Male, 0 = Female
}

df = pd.DataFrame(data)
print(df)
```

output

```
•••     Height  Weight  Age  Gender
    0     170      65   30       1
    1     165      59   25       0
    2     180      75   35       1
    3     175      68   28       1
    4     160      55   22       0
    5     172      70   32       1
    6     168      62   27       0
    7     177      74   33       1
    8     162      58   24       0
    9     158      54   21       0
```

```
X = df.drop('Gender', axis=1)
```

y = df['Gender']

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

X_train, X_test, y_train, y_test = train_test_split(X_pca, y, test_size=0.3, random_state=42)

model = LogisticRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

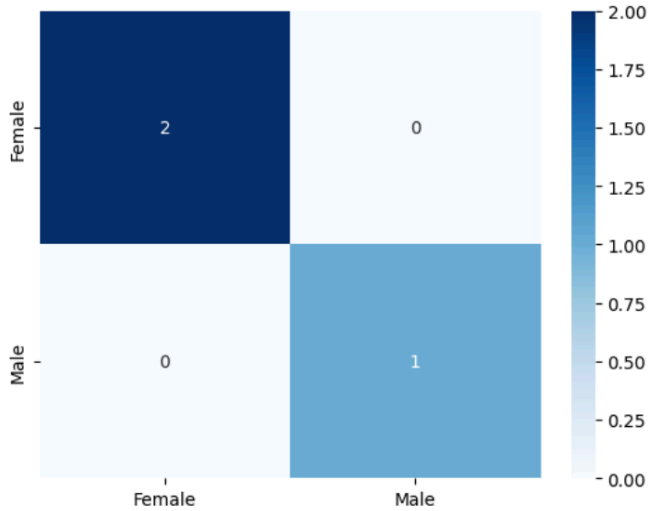
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Female', 'Male'], yticklabels=['Female', 'Male'])

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Female', 'Male'], yticklabels=['Female', 'Male'])
```

```
<Axes: >
```

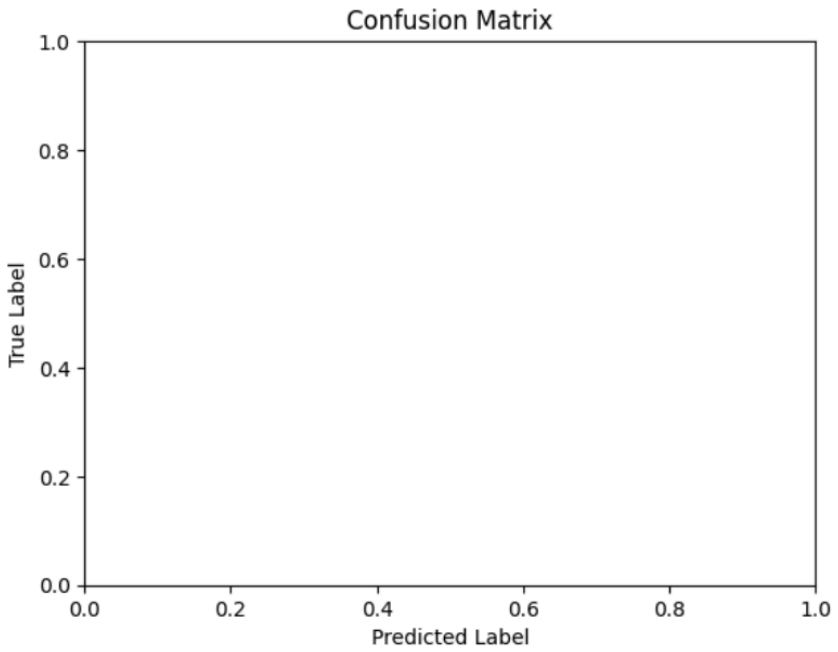


plt.xlabel('Predicted Label')

plt.ylabel('True Label')

plt.title('Confusion Matrix')

plt.show()



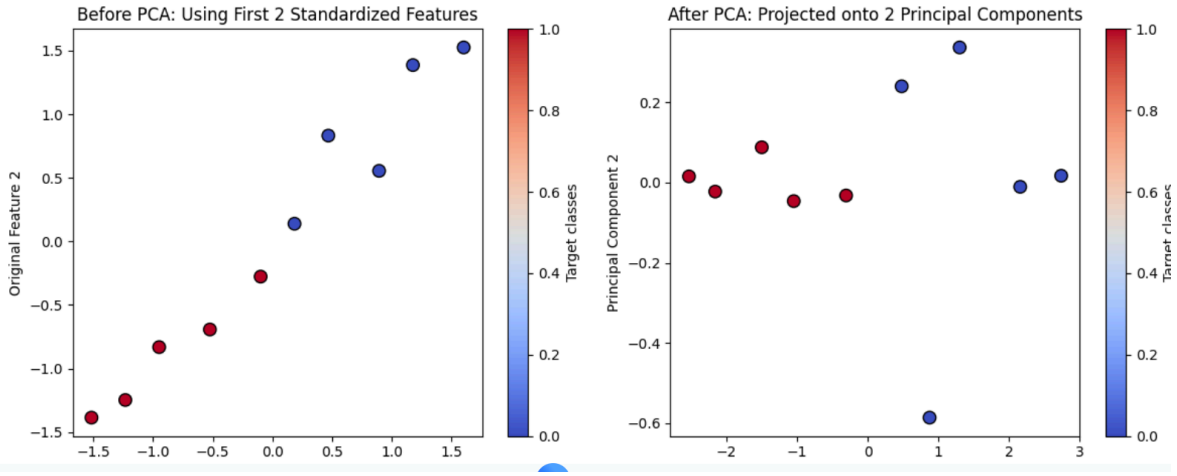

```
y_numeric = pd.factorize(y)[0]


plt.figure(figsize=(12, 5))


plt.subplot(1, 2, 1)
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_numeric, cmap='coolwarm', edgecolor='k',
s=80)
plt.xlabel('Original Feature 1')
plt.ylabel('Original Feature 2')
plt.title('Before PCA: Using First 2 Standardized Features')
plt.colorbar(label='Target classes')


plt.subplot(1, 2, 2)
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y_numeric, cmap='coolwarm', edgecolor='k', s=80)
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('After PCA: Projected onto 2 Principal Components')
plt.colorbar(label='Target classes')
```

```
plt.tight_layout()

plt.show()
```

...